

OXYRA: AN LLM-BASED CHATBOT AS AN INTELLIGENT AIR QUALITY INFORMATION SYSTEM

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Abstract— Air quality information is widely available, yet it is generally presented as static numerical indices that the general public finds difficult to interpret, and therefore fails to encourage preventive action. Large Language Models (LLMs) offer a natural-language interface that could bridge this gap, but they introduce the risk of hallucination, namely the delivery of convincing information that is not supported by data—a hazardous property in the public-health domain. This research develops OXYRA, an LLM-based chatbot that simplifies air quality information while suppressing hallucination through a separation of roles: all numerical values are retrieved by program code through a tool-calling mechanism, health recommendations are determined deterministically from the US EPA AQI categories, and the language model is restricted to composing the answer. A layered anti-hallucination strategy is applied, consisting of an out-of-scope question gate, region validation, and compound-question decomposition. The same architecture is deployed on two subsystems with different data characteristics: real-time data from 16 cities in East Java (IQAir API) and historical US EPA data whose AQI values are computed independently from raw concentrations. Testing employed 40 questions collected from lay respondents. With Llama 3.1 8B as the core model, the system obtained an average expert score of 8.58 on a 5–10 scale, compared to 8.25 for the comparison model Qwen3 8B. The core model produced no answer claiming a condition unsupported by data, whereas the comparison model produced six. Validation of 18 reference-query questions on the US EPA subsystem showed full agreement with the database. All hallucinations of the comparison model occurred when the model did not invoke a function, identifying compliance with the tool-calling mechanism as a determinant of system reliability.

Index Terms— air quality, Air Quality Index, chatbot, hallucination, Large Language Model, tool calling.

I. INTRODUCTION

AIR pollution has become one of the most pressing environmental issues in Indonesian cities. According to the WHO Global Air Quality Guidelines, prolonged exposure to fine particulate matter is associated with elevated risks of cardiovascular and respiratory diseases [5], and the health impacts of air pollution have been reviewed extensively in the literature [6].

Several studies have attempted to map urban air quality through sensor networks and mobile monitoring, such as PM_{2.5} mapping in Jakarta using mobile monitoring [7], the development of dense sensor networks for air-quality data validation [8], and the multi-sensor mobile device deployed

at street corners in the SMILE project [9]. Open-data assimilation across pan-European cities has also been proposed to address the limited coverage of official monitoring stations [10]. Despite their contributions, these efforts focus primarily on data acquisition and visualization, and have yet to bridge the gap between numerical data and public understanding. Notably, mobile monitoring in Jakarta demonstrated that pollutant concentrations can vary substantially within a single city [7], which implies that information derived from a single monitoring point must be communicated as a city-scale indication rather than as a value valid at a user's specific location.

Advances in Large Language Models (LLMs) open opportunities for more intuitive natural-language interfaces. Modern LLMs are built on the Transformer architecture introduced by Vaswani et al. [11], whose scaling gave rise to models capable of few-shot learning without task-specific training, as demonstrated by Brown et al. [12]; a comprehensive survey of this model family and its capabilities is provided by Zhao et al. [13]. Caldarini et al. surveyed recent advances in chatbots [1], while Kooli critically examined the use of chatbots in educational settings [2]. Yang et al. demonstrated the effectiveness of LLMs in handling tabular data [3], and VayuBuddy by Patel et al. provided a concrete example of using an LLM to democratize air-quality information in India through dynamic code execution [4].

The application of LLMs to public-health information, however, is constrained by hallucination. Ji et al. defined hallucination as model output unsupported by its source [14], and Huang et al. subsequently established a taxonomy showing that hallucination cannot be eliminated by model training alone [15]. In the context of air quality, a fabricated index value may mislead a user's decision to exercise outdoors or to wear protective equipment. Grounding the model in an external data source is therefore essential. Two grounding approaches are commonly used: Retrieval-Augmented Generation [16], and function calling, pioneered by Toolformer [17] and surveyed more broadly by Qin et al. [18].

This research develops OXYRA, an LLM-based chatbot for air quality information. Unlike VayuBuddy, which allows

the model to generate the code that queries the data, OXYRA provides a fixed set of predefined functions: the model only selects a function and fills in its parameters, while all database queries are written in program code. This design deliberately trades a degree of flexibility for guaranteed traceability, on the grounds that the domain served concerns public health, where a single fabricated value may mislead action.

The objectives of this research are threefold: to develop an LLM-based chatbot as an interactive and contextual medium for delivering air quality information; to provide relevant and comprehensible preventive recommendations based on real-time air quality conditions; and to implement a hallucination-control mechanism ensuring that all factual information delivered is grounded in the data the system holds.

II. SYSTEM DESIGN

A. System Architecture

The OXYRA architecture consists of three layers. The frontend is built with React and TypeScript, and is responsible for receiving questions, displaying responses, and rendering chart visualizations. The backend, built with FastAPI, acts as the orchestrator: it applies the anti-hallucination gates, facilitates function calling by the language model, retrieves the relevant data, and assembles the context returned to the model. The language-model layer runs locally through Ollama, so that inference does not depend on paid cloud services and user data privacy is preserved.

The governing principle of the architecture is a separation of roles: program code retrieves and computes the data, while the language model only composes the sentence. The model never produces a numerical value on its own; every figure originates from the execution of a function.

B. Data Sources and Subsystems

The system is deployed on two subsystems with deliberately different data characteristics, so that the generality of the approach can be assessed.

The real-time subsystem covers 16 cities in East Java, with data obtained from the IQAir API. A Python-based collector runs every two hours on a Virtual Private Server, scheduled through cron, while the data-serving API runs permanently as a systemd service. Each reading yields the US AQI value, the dominant pollutant code, and weather parameters. The community tier of the API does not provide raw pollutant concentrations, and the collection interval is set to two hours to remain within the monthly call quota.

The historical subsystem covers US EPA measurement data for four gaseous pollutants—ozone (O₃), nitrogen dioxide (NO₂), sulfur dioxide (SO₂), and carbon monoxide (CO)—across 53 states. In contrast to the real-time subsystem, this data provides raw concentrations without an index, so the AQI is computed by the system itself using the EPA piecewise linear interpolation [19]:

$$I_p = ((I_{Hi} - I_{Lo}) / (BP_{Hi} - BP_{Lo})) \times (C_p - BP_{Lo})$$

$$+ I_{Lo} \quad (1)$$

where I_p is the index for pollutant p , C_p is the rounded concentration, BP_{Lo} and BP_{Hi} are the concentration breakpoints bounding C_p , and I_{Lo} and I_{Hi} are the corresponding index values. The final AQI of a location is the maximum sub-index among all pollutants computed.

Applying an identical architecture to both subsystems tests whether the approach depends on a particular data source.

C. Tool Calling Mechanism

The language model is given a set of function definitions, each with a name, a description, and typed parameters. Upon receiving a question, the model selects the appropriate function and supplies its arguments; the program code then executes the function to retrieve data. The functions on the real-time subsystem cover single-city conditions, all-city conditions, inter-city ranking, and historical pattern analysis. The functions on the US EPA subsystem additionally support pollutant-specific queries, inter-state and inter-county comparison, and date-filtered retrieval, requiring the model to extract a temporal entity from natural language and convert it into a database-compatible format.

Health recommendations are not delegated to the language model. They are determined deterministically by code from the AQI category, paraphrasing the official US EPA cautionary statements; the model merely renders them into accessible language without altering their substance.

D. Layered Anti-Hallucination Strategy

Five layers operate in sequence before an answer is produced:

1. Numerical separation. All values originate from function execution; the model composes words only.
2. Out-of-scope gate. Questions concerning data the system does not hold—oxygen levels, $\mu\text{g}/\text{m}^3$ concentrations, pollution percentages, future forecasts, indoor air quality, or spatial policy—are politely declined before reaching the model.
3. Region validation. When a user refers to an unspecified location (e.g., "my area"), the system requests the city name rather than inventing one.
4. Compound-question decomposition. Questions containing multiple intents are decomposed; answerable parts are processed and out-of-scope parts declined, then merged into a single coherent answer.
5. Data-failure handling. When requested data is unavailable, the system states so honestly rather than fabricating a condition.

The strategy is deliberately conservative: the system prefers to decline when in doubt. For public-health information, an honest refusal is considered safer than a plausible but unsupported figure.

E. Language Models

Two locally-run models of comparable size were evaluated on an identical pipeline: Llama 3.1 8B [20] as the core

model, and Qwen3 8B [21] as the comparison model. Model selection was constrained by a technical prerequisite: support for the tool-calling mechanism. SahabatAI 8B, which excels in Indonesian-language nuance, was initially considered but could not be used because it does not support function calling—a finding that itself demonstrates how tool-calling support restricts model choice in this class of architecture.

III. TESTING METHODOLOGY

A. Accuracy Validation — US EPA Subsystem

The US EPA subsystem serves as a technical accuracy and standards-compliance test rather than a perception test. It is the only subsystem for which a local database of raw concentrations is available, so that the values delivered by the system can be re-examined through a query, the category verified against the EPA breakpoints, and the recommendation verified against the EPA cautionary statements. All three references are deterministic and published, allowing the results to be independently reproduced.

Thirty validation questions were composed by the researcher to ensure that the test covers every pollutant and every query type supported by the system; these questions are not intended to represent lay phrasing. Reporting focuses on the 18 questions that have a database reference query—13 technical and 5 applied.

B. Lay-Perception Testing — East Java Subsystem

Forty questions were collected from lay respondents through an online questionnaire, with respondents asked to phrase their questions as if asking a friend. All 40 questions were submitted to the system without filtering, including out-of-scope and off-topic questions, so that system behavior would be recorded under realistic conditions.

Twelve questions were then selected for expert assessment using purposive sampling, with the criterion that the selection must collectively represent every processing path of the system: current-condition queries, educational questions, unspecified locations, out-of-scope questions, and compound questions. Random sampling was not used, because it risks producing a homogeneous sample in which the anti-hallucination layer—the principal contribution of this work—would go untested. The purpose of expert assessment here is representational coverage of system functions, not statistical generalization. The selected answers were scored on a 5–10 scale by an independent expert validator in environmental engineering, Prof. Ir. Joni Hermana, M.Sc.ES., Ph.D., who is not an author of this work and was not involved in the design or implementation of the system..

C. Comparative Model Testing

The two models were compared on an identical pipeline: all backend components, tools, anti-hallucination gates, and data were held constant, and only the language model was swapped. On the East Java subsystem, the 40 original respondent questions were used verbatim, including non-standard spelling. On the

US EPA subsystem, the same 40 questions were used with respondent wording preserved and only the location entities adjusted (e.g., Surabaya to California), so that linguistic style is controlled and only the data context varies.

Two interpretive adjustments must be noted. First, temporal semantics shift between subsystems: "today" refers to a real-time condition on the East Java subsystem, but to the latest available record on the historical subsystem. Second, data coverage differs: the East Java subsystem holds weather parameters while the US EPA subsystem does not, so temperature-related questions are declined by the out-of-scope gate on the latter. Such behavioral differences arise from data coverage rather than architectural failure; consistent refusal in fact demonstrates that the gate operates as designed.

D. Evaluation Metrics

Three metrics were used. Expert score (5–10 scale) was obtained from the 12 assessed questions on the East Java subsystem. Hallucination count and response time were measured objectively across all 80 answers, without requiring expert judgment, and are therefore reproducible from the test records.

Hallucination is defined operationally so that it can be re-audited: an answer is deemed hallucinated if it claims an actual condition—an AQI value, temperature, humidity, dominant pollutant, or location—that does not appear in the recorded function-call data for that question. Mentioning a figure while explaining the AQI scale—for instance, stating that values above 100 are unhealthy—is not counted, as this is correct EPA standard knowledge rather than a claim about an actual condition.

IV. RESULTS AND DISCUSSION

A. Accuracy Validation

All 18 reference-query questions on the US EPA subsystem were answered with values and entities matching the reference query results (18 of 18). Table 1 summarizes the distribution.

Table 1.
Accuracy validation against database reference queries

Category	Reference form	Count	Matched
Technical	Pollutant values, computed AQI, inter-region ranking, hourly patterns	13	13
Applied	Location and time with the best AQI in the requested range	5	5
Total		18	18

This result is consistent with the architectural principle: all numerical values originate from code computation rather than from the language model, so numerical error is reduced to zero in this test. It further demonstrates that the independent AQI computation from raw concentrations conforms to the EPA standard, since the values computed by the system agree with those computed directly from the database.

A representative case illustrates all three verification layers at once. Asked about air quality in Los Angeles on a specific date, the system correctly extracted the temporal parameter from natural language and passed it to the function call; every concentration value it reported was present in the corresponding data rows; and its computed AQI was consistent with the EPA breakpoints, with an ozone concentration of 0.066 ppm yielding an AQI of 87 in the Moderate category while other stations remained in the Good category.

B. Expert Assessment

With Llama 3.1 8B as the core model, the system obtained an average expert score of 8.58 on the 5–10 scale, compared with 8.25 for Qwen3 8B.

The composition of the assessed sample is worth noting: of the 12 questions, only two could be answered through the normal path. The remainder tested the anti-hallucination layer—four concerned data the system does not hold, four referred to locations without specification, and three contained multiple intents. The score is therefore conservative, reflecting the system's performance on challenging rather than favorable questions. Polite refusals of out-of-scope questions were rated as good answers by the expert, not as system failures.

C. Comparative Model Testing

Table 2 summarizes the comparative results.

Table 2.
Comparative results of the two models on an identical pipeline

Aspect	Llama 3.1 8B	Qwen3 8B
Expert score, 5–10 scale (12 questions)	8.58	8.25
Answers claiming conditions without data support	0	6
Mean response time, East Java (s)	15.4	31.4
Mean response time, US EPA (s)	24.4	36.5

Each model answered the same 80 questions (40 per subsystem), producing 80 answers per model. An audit of these answers found six Qwen3 answers claiming conditions without data support, all of them on the East Java subsystem, while every Llama 3.1 answer passed the audit without a single fabricated value.

The failure pattern is consistent and consequential: all six Qwen3 hallucinations occurred in answers whose function-call record was empty—that is, when the model chose not to invoke any tool and instead composed an answer from its own parametric knowledge. Three of these answers

presented a list of AQI values for several cities, none of which originated from the system's data; two fabricated an AQI value together with temperature and humidity; and one referred to a city that the system does not monitor at all.

Notably, no hallucination was recorded on the US EPA subsystem for either model. On that subsystem, a question that does not specify a pollutant or a sufficiently narrow region is intercepted by the clarification gate, so the model has no opportunity to improvise. The structure of the data, which compels specificity, therefore acts as an additional safeguard.

This finding delineates the operating boundary of the anti-hallucination architecture. The filtering gates and numerical separation function as designed, but the guarantee holds only insofar as the model complies with the function-calling mechanism. Compliance with tool calling is therefore not merely a technical prerequisite but a determinant of system reliability—and it is precisely on this dimension that the two models diverge sharply. The distinction is decisive: in a public-health information system, a single fabricated value is more damaging than a difference in linguistic style.

The expert scores of the two models are relatively close (8.58 against 8.25), indicating that both produce language of acceptable quality. The assessor also noted a stylistic difference: Llama 3.1 answers were judged more fluent and accessible to lay users, whereas Qwen3 tended to be more structured and data-oriented. In response time, Llama 3.1 was consistently about twice as fast on the East Java subsystem and about one and a half times faster on the US EPA subsystem; part of this margin arises from Qwen3's internal reasoning mode, which produces a deliberation trace before the final answer.

The combination of these three aspects—freedom from hallucination, faster response, and a slightly higher expert score—confirms Llama 3.1 8B as the core model of OXYRA.

D. Architectural Contribution

To isolate the contribution of the architecture from that of the model, an identical question was posed to the same base model (Llama 3.1 8B) run bare through Ollama and run within the OXYRA system.

Run bare, the model produced a lengthy answer containing claims unsupported by measured data: it asserted an air-quality condition without accessing any data, speculated about medical diagnoses, and offered recommendations extending well beyond the domain, including hygiene and diet. Run within OXYRA, the same model produced a concise answer grounded in the data the system holds—current AQI, temperature, humidity, and dominant pollutant obtained through function calls—with a qualitative and bounded explanation of the health connection. Table 3 summarizes the comparison.

Table 3.
Bare model versus OXYRA system (identical base model)

Dimension	Bare LLM (Ollama)	Through OXYRA
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Information source	Parametric knowledge, no data access	Actual data via function calls
Air-quality claim	Qualitative, unfounded, contradicting actual conditions	AQI, temperature, humidity per data
Recommendation scope	Extends into medical and lifestyle domains	Confined to air quality
Traceability	Not auditable	Every figure traceable to its source

E. Discussion

The results position this work relative to prior studies in three respects.

First, the survey by Caldarini et al. [1] observed that the flexibility of modern chatbots comes at the cost of answer reliability. The present results provide empirical evidence for this tension and a means of managing it: reliability is recovered not by curtailing the model's linguistic flexibility, but by confining its role to sentence composition.

Second, Yang et al. [3] established the feasibility of LLMs in processing structured data. This work refines that finding: feasibility is not uniform across models. Qwen3 8B, despite supporting function calling, elected on several occasions not to invoke a function at all and composed answers from parametric knowledge, producing values absent from the data. Feasibility therefore depends not only on a model's capacity to comprehend data, but on its compliance in using the retrieval path provided.

Third, this work diverges from VayuBuddy [4] at a decisive point. VayuBuddy allows the model to generate the code that queries the data, maximizing flexibility but making the guarantee contingent on the correctness of the generated code. OXYRA supplies a fixed set of functions, so that every value delivered is necessarily the product of code execution. The trade-off is explicit: questions not anticipated in the function design cannot be answered, and the system declines them honestly rather than forcing a response.

V. CONCLUSION

This research designed, implemented, and evaluated OXYRA, an LLM-based chatbot that simplifies air quality information for lay users while suppressing hallucination through a separation of roles between program code and the language model.

Expert assessment of twelve representative questions drawn from 40 lay respondents yielded an average score of 8.58 on a 5–10 scale, indicating that conversational delivery can overcome the limitations of static numerical presentation. Health recommendations were determined deterministically from the US EPA AQI categories, addressing the finding that most lay questions concern implications for action rather than technical values. An audit of the 80 answers produced by each

model showed that the core model, Llama 3.1 8B, reported no condition unsupported by data, whereas the comparison model did so in six answers, and validation of 18 reference-query questions on the US EPA subsystem showed full agreement with the database.

Two further findings emerged. Compliance with the tool-calling mechanism proved to be a determinant of system reliability: every hallucination observed in the comparison model occurred in answers that did not invoke a function. And the same architecture served two markedly different data characteristics—a pre-computed real-time index and raw historical concentrations—without modification to its core mechanism, indicating that the approach is not bound to a particular data source.

Several limitations remain. Expert assessment covered twelve questions and a single independent assessor, who was not involved in the design or implementation of the system; the out-of-scope gate relies on keyword matching and is deliberately conservative; each city is represented by a single monitoring point; and the 8B-class model exhibits variation in sentence composition and in its decision to invoke functions.

Future work may integrate curated online sources through a retrieval mechanism to address needs beyond the system's data coverage, provided that a curation layer preserves the anti-hallucination guarantee; refine intent detection through machine-learning classification; increase monitoring-point density [8], [9]; expand the number of assessed questions and assessors; and evaluate larger locally-deployable models.

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